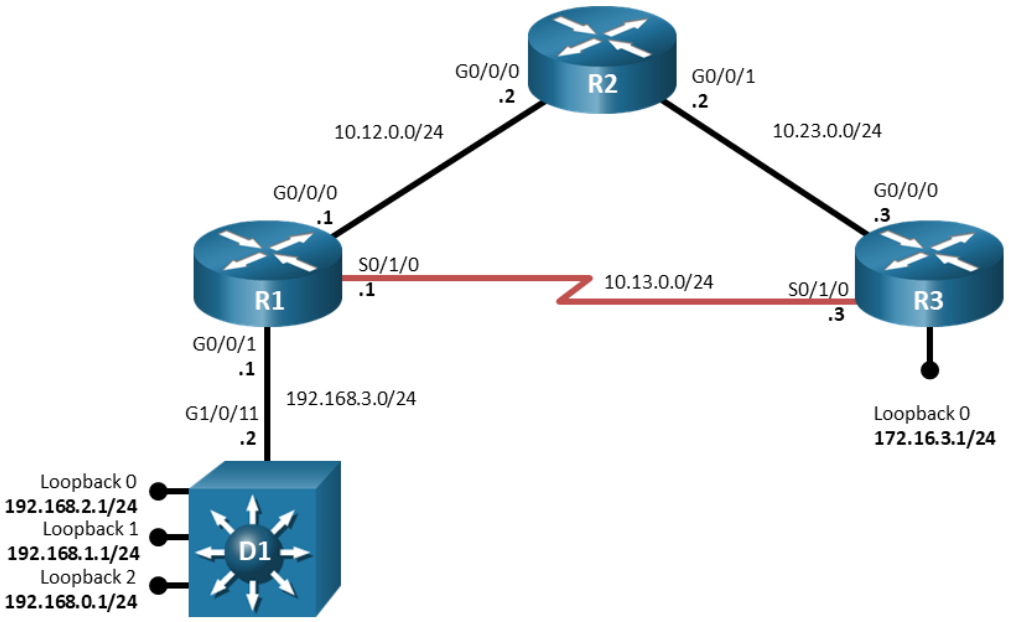
Lab - Implement Advanced EIGRP for IPv4 Features

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask
R1	G0/0/0	10.12.0.1	255.255.255.0
	G0/0/1	192.168.3.1	255.255.255.0
	S0/1/0	10.13.0.1	255.255.255.0
R2	G0/0/0	10.12.0.2	255.255.255.0
	G0/0/1	10.23.0.2	255.255.255.0
R3	G0/0/0	10.23.0.3	255.255.255.0
	S0/1/0	10.13.0.3	255.255.255.0
	Loopback0	172.16.3.1	255.255.255.0
D1	G1/0/11	192.168.3.2	255.255.255.0
	Loopback0	192.168.1.1	255.255.255.0
	Loopback1	192.168.1.1	255.255.255.0
	Loopback2	192.168.0.1	255.255.255.0

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Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Implement EIGRP for IPv4

Part 3: Implement Advanced Features

Background / Scenario

Customizing the operation of EIGRP can yield many benefits, most notably speeding convergence and stabilizing network operations during outages. In this lab you will explore some advanced techniques that can be used to customize and improve EIGRP performance on an enterprise network.

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Make sure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 1 Switch (Cisco 9300L with Cisco IOS XE release 17.12.5 universal image or comparable)
- 1 PC (Choice of operating system with a terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings and interface addressing.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Router R1

```
hostname R1
no ip domain lookup
banner motd # R1, Implement Advanced EIGRP for IPv4 Features #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
```

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```
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface g0/0/0
 ip address 10.12.0.1 255.255.255.0
 no shutdown
 exit
interface s0/1/0
 ip address 10.13.0.1 255.255.255.0
 no shutdown
 exit
interface g0/0/1
 ip address 192.168.3.1 255.255.255.0
 no shutdown
 exit
end
```

Router R2

```
hostname R2
no ip domain lookup
banner motd # R2, Implement Advanced EIGRP for IPv4 Features #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface g0/0/0
 ip address 10.12.0.2 255.255.255.0
 no shutdown
 exit
interface g0/0/1
 ip address 10.23.0.2 255.255.255.0
 no shutdown
 exit
end
```

Router R3

```
hostname R3
```

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```
no ip domain lookup
banner motd # R3, Implement Advanced EIGRP for IPv4 Features #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 transport input telnet
 exit
interface g0/0/0
 ip address 10.23.0.3 255.255.255.0
 no shutdown
 exit
interface s0/1/0
 ip address 10.13.0.3 255.255.255.0
 no shutdown
 exit
interface loopback 0
 ip address 172.16.3.1 255.255.255.0
 no shutdown
 exit
end
```

Switch D1

```
hostname D1
no ip domain lookup
ip routing
banner motd # D1, Implement Advanced EIGRP for IPv4 Features #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface range g1/0/1-24
 shutdown
 exit
interface g1/0/11
```

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```
 no switchport
 ip address 192.168.3.2 255.255.255.0
 no shutdown
 exit
interface loopback 0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit
interface loopback 1
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit
interface loopback 2
 ip address 192.168.0.1 255.255.255.0
 no shutdown
 exit
end
```

b. Set the clock on all devices to UTC time.

c. Save the running configuration to startup-config on all devices.

Part 2: Implement EIGRP for IPv4

In this part, you will configure classic EIGRP for IPv4 and verify that all routing tables are converged.

Step 1: Configure classic EIGRP for IPv4.

a. Enable debugging on R1 so you can observe the route exchange process.

```
R1# debug eigrp packets
    (UPDATE, REQUEST, QUERY, REPLY, HELLO, UNKNOWN, PROBE, ACK, STUB,
SIAQUERY, SIAREPLY)
EIGRP Packet debugging is on
```

b. Configure classic EIGRP for IPv4 on all devices. Use Autonomous System number 98 and advertise only the connected interfaces on each device.

Step 2: Review the debugging results.

a. Stop debugging.

```
R1# no debug eigrp packets
```

b. Review the debugging results. As soon as you enable EIGRP on R1, EIGRP starts to send and receive hello packets for neighbor discovery. In this example, when R1 receives hello packets from the neighbors R2 and D1, R1 forms new EIGRP adjacencies.

Update packets with routing information are sent to the newly discovered neighbors. When there are link or metric changes, the update packets are multicast to the neighbors.

EIGRP sends query packets to find feasible successors to a destination from a neighbor router. The neighbor router sends a reply packet providing feasible successor information in response to the sender of the query packet.

Acknowledgment (ACK) packets confirm the successful receipt of EIGRP update, query or reply packets.

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```
*Oct 17 01:16:54.040: EIGRP: Sending HELLO on Gi0/0/0 - paklen 20
*Oct 17 01:16:54.040:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidbQ un/rely
0/0
*Oct 17 01:16:54.044: EIGRP: Sending HELLO on Se0/1/0 - paklen 20
*Oct 17 01:16:54.044:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidbQ un/rely
0/0
*Oct 17 01:16:54.044: EIGRP: Sending HELLO on Gi0/0/1 - paklen 20
*Oct 17 01:16:54.044:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidbQ un/rely
0/0
*Oct 17 01:16:54.085: EIGRP: Received HELLO on Gi0/0/0 - paklen 30 nbr 10.12.0.2
*Oct 17 01:16:54.095:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0
*Oct 17 01:16:54.104: EIGRP: Add Peer: Total 1 (1/0/0/0/0)
*Oct 17 01:16:54.104: EIGRP: Add Peer: Total 1 (1/0/1/0/0)
*Oct 17 01:16:54.104: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.12.0.2
(GigabitEthernet0/0/0) is up: new adjacency
*Oct 17 01:16:54.113: EIGRP: Enqueueing UPDATE on Gi0/0/0 - paklen 0 nbr 10.12.0.2 tid
0 iidbQ un/rely 0/1 peerQ un/rely 0/0
*Oct 17 01:16:54.114: EIGRP: Received HELLO on Gi0/0/1 - paklen 30 nbr 192.168.3.2
*Oct 17 01:16:54.114:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0
*Oct 17 01:16:54.114: EIGRP: Add Peer: Total 1 (1/0/0/0/0)
*Oct 17 01:16:54.114: EIGRP: Add Peer: Total 1 (1/0/1/0/0)
*Oct 17 01:16:54.114: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 192.168.3.2
(GigabitEthernet0/0/1) is up: new adjacency
*Oct 17 01:16:54.114: EIGRP: Enqueueing UPDATE on Gi0/0/1 - paklen 0 nbr 192.168.3.2
tid 0 iidbQ un/rely 0/1 peerQ un/rely 0/0
*Oct 17 01:16:54.114: EIGRP: Received HELLO on Se0/1/0 - paklen 30 nbr 10.13.0.3
*Oct 17 01:16:54.114:   AS 98, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0
*Oct 17 01:16:54.114: EIGRP: Add Peer: Total 1 (1/0/0/0/0)
*Oct 17 01:16:54.114: EIGRP: Add Peer: Total 1 (1/0/1/0/0)
*Oct 17 01:16:54.114: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.13.0.3
(Serial0/1/0) is up: new adjacency
*Oct 17 01:16:54.114: EIGRP: Enqueueing UPDATE on Se0/1/0 - paklen 0 nbr 10.13.0.3 tid
0 iidbQ un/rely 0/1 peerQ un/rely 0/0
*Oct 17 01:16:54.114: EIGRP: Received UPDATE on Se0/1/0 - paklen 0 nbr 10.13.0.3
*Oct 17 01:16:54.114:   AS 98, Flags 0x1:(INIT), Seq 5/0 interfaceQ 0/0 iidbQ un/rely
0/1 peerQ un/rely 0/0
<output omitted>
*Oct 17 01:16:54.160: EIGRP: Requeued unicast on GigabitEthernet0/0/1
*Oct 17 01:16:54.160: EIGRP: Requeued unicast on Serial0/1/0
*Oct 17 01:16:54.160: EIGRP: Sending UPDATE on Gi0/0/0 - paklen 0 nbr 10.12.0.2 tid 0
*Oct 17 01:16:54.160:   AS 98, Flags 0x1:(INIT), Seq 1/0 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 0/1
*Oct 17 01:16:54.168: EIGRP: Sending UPDATE on Gi0/0/1 - paklen 0 nbr 192.168.3.2 tid
0
*Oct 17 01:16:54.168:   AS 98, Flags 0x1:(INIT), Seq 2/0 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 0/1
*Oct 17 01:16:54.168: EIGRP: Sending UPDATE on Se0/1/0 - paklen 0 nbr 10.13.0.3 tid 0
*Oct 17 01:16:54.168:   AS 98, Flags 0x1:(INIT), Seq 3/5 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 0/1
*Oct 17 01:16:54.171: EIGRP: Received ACK on Se0/1/0 - paklen 0 nbr 10.13.0.3
*Oct 17 01:16:54.171:   AS 98, Flags 0x0:(NULL), Seq 0/3 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 0/1
```

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```
*Oct 17 01:16:54.171: EIGRP: Enqueueing UPDATE on Se0/1/0 - paklen 0 tid 0 iidbQ
un/rely 0/1 serno 1-3
*Oct 17 01:16:54.172: EIGRP: Enqueueing UPDATE on Se0/1/0 - paklen 0 nbr 10.13.0.3 tid
0 iidbQ un/rely 0/0 peerQ un/rely 0/0 serno 1-3
*Oct 17 01:16:54.183: EIGRP: Received UPDATE on Se0/1/0 - paklen 132 nbr 10.13.0.3
*Oct 17 01:16:54.183:   AS 98, Flags 0x8:(EOT), Seq 6/3 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 0/1
*Oct 17 01:16:54.183: EIGRP: Enqueueing ACK on Se0/1/0 - paklen 0 nbr 10.13.0.3 tid 0
*Oct 17 01:16:54.183:   Ack seq 6 iidbQ un/rely 0/0 peerQ un/rely 1/1
*Oct 17 01:16:54.214: EIGRP: Sending ACK on Se0/1/0 - paklen 0 nbr 10.13.0.3 tid 0
*Oct 17 01:16:54.214:   AS 98, Flags 0x0:(NULL), Seq 0/6 interfaceQ 0/0 iidbQ un/rely
0/0 peerQ un/rely 1/1
*Oct 17 01:16:54.214: EIGRP: Sending UPDATE on Se0/1/0 - paklen 88 nbr 10.13.0.3 tid 0
```

Step 3: Verify EIGRP for IPv4 routing.

a. Verify that each device has a complete routing table for all the networks shown in the topology and Addressing Table.

b. Issue the **show ip eigrp traffic** command to verify EIGRP packet traffic statistics on AS 98. This command provides statistical information about the types of packets sent and received by the devices.

```
R2# show ip eigrp traffic
EIGRP-IPv4 Traffic Statistics for AS(98)
  Hellos sent/received: 649/646
  Updates sent/received: 16/13
  Queries sent/received: 0/0
  Replies sent/received: 0/0
  Acks sent/received: 8/16
  SIA-Queries sent/received: 0/0
  SIA-Replies sent/received: 0/0
  Hello Process ID: 468
  PDM Process ID: 428
  Socket Queue: 0/10000/1/0 (current/max/highest/drops)
  Input Queue: 0/10000/1/0 (current/max/highest/drops)
```

Step 4: Choosing the best path.

EIGRP uses the Diffusing Update Algorithm (DUAL) to calculate the best paths to the destination networks. DUAL ensures loop-free routes to the destination networks and synchronizes simultaneously when there is a change in the topology. The network devices unaffected by the topology or link changes do not participate in the recalculations.

DUAL uses the distance information to determine the best route to the destination.

Reported or advertised distance (RD or AD): The metric (distance) to a specific destination network from the neighbor router as advertised by the neighboring router

Feasible distance (FD): The metric (total distance) of the best route to a specific destination network from the local router. The metric of the best route is the sum of the AD from the neighbor router and the metric to the neighbor router from the local router.

Successor Route: The route with the smallest metric

Successor: The successor has the best path the destination. Up to four successors can be displayed in the routing table by default.

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Feasible successor route: This route requires the RD of alternative backup route to be smaller than the FD of the best route.

Feasible successor: The next hop routers that satisfy the feature successor condition has the backup path to the destination.

For example, the FD to 172.16.3.0 on R3 is 131072 as noted in the routing table on R1. The best route to the destination networks on R3 is via G0/0/0 on R1.

```
R1# show ip route eigrp | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D        10.23.0.0/24 [90/3072] via 10.12.0.2, 00:06:05, GigabitEthernet0/0/0
      172.16.0.0/24 is subnetted, 1 subnets
D        172.16.3.0 [90/131072] via 10.12.0.2, 00:06:04, GigabitEthernet0/0/0
D     192.168.0.0/24
           [90/130816] via 192.168.3.2, 00:05:37, GigabitEthernet0/0/1
D     192.168.1.0/24
           [90/130816] via 192.168.3.2, 00:05:37, GigabitEthernet0/0/1
D     192.168.2.0/24
           [90/130816] via 192.168.3.2, 00:05:37, GigabitEthernet0/0/1
```

To review the alternate routes to the destination networks on R3, you can use the **show ip eigrp topology** commands.

a. On R1, enter the **show ip eigrp topology** command. Note the FD for to the remote network 172.16.3.0/24. Furthermore, note the alternate route to the destination via Serial0/1/0.

   The successor route to 172.16.3.0/24 is through G0/0/0 and FD is 131072, as displayed in the routing table. The advertised distance is 130816, which is smaller than FD, meets the requirements for a loop-free topology.

```
R1# show ip eigrp topology
EIGRP-IPv4 Topology Table for AS(98)/ID(192.168.3.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 192.168.3.0/24, 1 successors, FD is 2816
        via Connected, GigabitEthernet0/0/1
P 192.168.2.0/24, 1 successors, FD is 130816
        via 192.168.3.2 (130816/128256), GigabitEthernet0/0/1
P 10.13.0.0/24, 1 successors, FD is 1792000
        via Connected, Serial0/1/0
        via 10.12.0.2 (1782272/1782016), GigabitEthernet0/0/0
P 192.168.0.0/24, 1 successors, FD is 130816
        via 192.168.3.2 (130816/128256), GigabitEthernet0/0/1
P 192.168.1.0/24, 1 successors, FD is 130816
        via 192.168.3.2 (130816/128256), GigabitEthernet0/0/1
P 172.16.3.0/24, 1 successors, FD is 131072
        via 10.12.0.2 (131072/130816), GigabitEthernet0/0/0
        via 10.13.0.3 (1920000/128256), Serial0/1/0
P 10.23.0.0/24, 1 successors, FD is 3072
        via 10.12.0.2 (3072/2816), GigabitEthernet0/0/0
        via 10.13.0.3 (1792256/2816), Serial0/1/0
```

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```
P 10.12.0.0/24, 1 successors, FD is 2816
        via Connected, GigabitEthernet0/0/0
```

As displayed in the results, you can also reach 172.16.3.0/24 via the Serial 0/1/0 interface if the route via G0/0/0 is down.

b. Now that you have confirmed that you have a feasible successor from R1 to 172.16.3.0/24 network on R3, you can observe the quick EIGRP convergence when a link fails in the successor route. Send 100,000 pings from R1 to the loopback 0 interface (172.16.3.1) on R3. Then simulate a link failure by shutting down the G0/0/0 interface on R2.

```
R1# ping 172.16.3.1 repeat 100000
!!!
```

```
        via 10.12.0.2 (131072/130816), GigabitEthernet0/0/0
        via 10.13.0.3 (1920000/128256), Serial0/1/0
P 10.23.0.0/24, 1 successors, FD is 3072, serno 40
        via 10.12.0.2 (3072/2816), GigabitEthernet0/0/0
        via 10.13.0.3 (1792256/2816), Serial0/1/0
P 10.12.0.0/24, 1 successors, FD is 2816, serno 39
        via Connected, GigabitEthernet0/0/0
        via 10.13.0.3 (1792512/3072), Serial0/1/0
```

Part 3: Implement Advanced Features

In this part of the lab, you will customize several different settings within EIGRP and see the impact of those changes on the network.

Step 1: Modify timers.

EIGRP uses standard **hello-interval** and **hold-time** timers based on the speed of the interface. If the interface speed is a T1 or less, hellos are sent every 60 seconds and the hold-time is set to 180 seconds. If the interface speed is greater than a T1, hellos are sent every 5 seconds and the **hold-time** is set to 15 seconds. These default times might not be appropriate for some network scenarios.

a. To see what the timers are set to, issue the **show ip eigrp interfaces detail** command.

```
R1# show ip eigrp interfaces detail
EIGRP-IPv4 Interfaces for AS(98)
                              Xmit Queue   PeerQ        Mean   Pacing Time   Multicast    Pending
Interface              Peers  Un/Reliable  Un/Reliable  SRTT   Un/Reliable   Flow Timer   Routes
Gi0/0/0                  0        0/0          0/0         0       0/0           50           0
  Hello-interval is 5, Hold-time is 15
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 5/1
  Hello's sent/expedited: 29/2
  Un/reliable mcasts: 0/5  Un/reliable ucasts: 7/3
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 2  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:

Gi0/0/1                  1        0/0          0/0         1       0/0           50           0
  Hello-interval is 5, Hold-time is 15
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 12/1
  Hello's sent/expedited: 387/2
  Un/reliable mcasts: 0/12  Un/reliable ucasts: 13/5
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 2  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:
```

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```
Se0/1/0                   1        0/0       0/0           1       0/12          50           0
  Hello-interval is 5, Hold-time is 15
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 4/0
  Hello's sent/expedited: 32/2
  Un/reliable mcasts: 0/0  Un/reliable ucasts: 6/5
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 0  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:
```

b. For this lab, the timers on R1 interface G0/0/0 and S0/1/0 need to be adjusted to send hellos every 10 seconds and establish a hold time of 30 seconds. EIGRP is unique in that each interface can have a customized **hello-interval** and **hold-time**. The times are not needed to match between the ends of a link. Change the timers using the **ip hello-interval eigrp** *ASN seconds* and the **ip hold-time eigrp** *ASN seconds* interface configuration commands.

```
R1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)# interface g0/0/0
R1(config-if)# ip hello-interval eigrp 98 10
R1(config-if)# ip hold-time eigrp 98 30
R1(config-if)# exit
R1(config)# interface s0/1/0
R1(config-if)# ip hello-interval eigrp 98 10
R1(config-if)# ip hold-time eigrp 98 30
R1(config-if)# end
```

c. To verify that the changes were made, check the output of the **show ip eigrp interfaces detail** command.

```
R1# show ip eigrp interfaces detail
EIGRP-IPv4 Interfaces for AS(98)
                            Xmit Queue   PeerQ        Mean   Pacing Time   Multicast    Pending
Interface              Peers  Un/Reliable  Un/Reliable  SRTT   Un/Reliable   Flow Timer   Routes
Gi0/0/0                  0        0/0       0/0           0       0/0           50           0
  Hello-interval is 10, Hold-time is 30
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 12/2
  Hello's sent/expedited: 381/3
  Un/reliable mcasts: 0/9  Un/reliable ucasts: 12/7
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 4  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:

Gi0/0/1                   1        0/0       0/0           1       0/0           50           0
  Hello-interval is 5, Hold-time is 15
```

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```
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 12/1
  Hello's sent/expedited: 387/2
  Un/reliable mcasts: 0/12  Un/reliable ucasts: 13/5
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 2  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:

Se0/1/0                  1        0/0       0/0           1       0/12          50           0
  Hello-interval is 10, Hold-time is 30
  Split-horizon is enabled
  Next xmit serial <none>
  Packetized sent/expedited: 8/0
  Hello's sent/expedited: 273/2
  Un/reliable mcasts: 0/0  Un/reliable ucasts: 8/11
  Mcast exceptions: 0  CR packets: 0  ACKs suppressed: 0
  Retransmissions sent: 0  Out-of-sequence rcvd: 0
  Topology-ids on interface - 0
  Authentication mode is not set
  Topologies advertised on this interface:  base
  Topologies not advertised on this interface:
```

Step 2: Create summarized routes in EIGRP.

Large routing tables take more memory and require more CPU time to process. Reducing the size of the routing table is advantageous in all network scenarios. EIGRP supports summarization of routes at any point in the network. There is no boundary router limitation like the limitations imposed in OSPF. However, in order for the summary route to be valid, EIGRP requires that some component of the summary route be in the routing table for the router doing the summarization.

a. Issue the **show ip route eigrp | begin Gateway** command on R3 and note the group of networks from the 192.168 range of addresses. R1 is advertising this contiguous block of networks individually, instead of sending a summary.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D        10.12.0.0/24 [90/3072] via 10.23.0.2, 00:06:35, GigabitEthernet0/0/0
D     192.168.0.0/24 [90/131328] via 10.23.0.2, 00:06:08, GigabitEthernet0/0/0
D     192.168.1.0/24 [90/131328] via 10.23.0.2, 00:06:08, GigabitEthernet0/0/0
D     192.168.2.0/24 [90/131328] via 10.23.0.2, 00:06:08, GigabitEthernet0/0/0
D     192.168.3.0/24 [90/3328] via 10.23.0.2, 00:06:35, GigabitEthernet0/0/0
```

b. On R1, configure a summary of the networks between R1 and D1, as well as the networks on D1 on the interfaces connecting to R3 and R2.

```
R1# conf t
R1(config)# interface g0/0/0
R1(config-if)# ip summary-address eigrp 98 192.168.0.0 255.255.252.0
```

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```
*Mar  6 00:07:33.742: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.12.0.2
(GigabitEthernet0/0/0) is resync: summary configured
R1(config-if)# exit
R1(config)# interface s0/1/0
R1(config-if)# ip summary-address eigrp 98 192.168.0.0 255.255.252.0
*Mar  6 00:07:35.220: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.13.0.3
(Serial0/1/0) is resync: summary configured
R1(config-if)# end
```

c. Now examine the routing table on R3 again using the **show ip route eigrp | begin Gateway** command. In the output, you now see a single route taking the place of what had been four distinct routes.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D        10.12.0.0/24 [90/3072] via 10.23.0.2, 00:07:59, GigabitEthernet0/0/0
D     192.168.0.0/22 [90/3328] via 10.23.0.2, 00:00:25, GigabitEthernet0/0/0
```

Step 3: Control EIGRP query propagation with EIGRP stub routers.

EIGRP uses query messages to find a path to networks in the autonomous system. The query messages always require an acknowledgement. But a router will only send a response if it has a potential route that satisfies the query. If it does not have a route, it sends its own queries to its neighbors. This process can lead to long delays in reconvergence after an outage.

Query scoping refers to using various techniques to control how far across a network queries have to be sent. Summarization is one way of controlling query propagation. Another way to control query propagation is to use EIGRP stub routers where appropriate. When a router is single-homed to the rest of the network, and no other networks exist beyond that router, there is no real point in sending it a query looking for lost networks. The stub router declares itself as a stub to the router connected to the rest of the network, which is considered a hub router. The hub router then forwards no queries to the stub router because it knows there are no other networks, beyond those reported, existing beyond the stub router. In the topology for this lab, switch D1 is a stub router and R1 is its hub router.

a. To verify that switch D1 is receiving EIGRP queries, issue the **shutdown** command on R2 interface G0/0/1. On switch D1, issue the **show ip eigrp events** command. This command outputs a timestamped list of actions that EIGRP is taking. In the output, you will find an entry that says switch D1 received a query trying to find the 10.23.0.0/24 network. Take note of the time stamp.

```
D1# show ip eigrp events
Event information for AS 98:
1    00:10:46.753 NDB delete: 10.23.0.0/24 1
2    00:10:46.753 RDB delete: 10.23.0.0/24 192.168.3.1
3    00:10:46.737 Metric set: 10.23.0.0/24 metric(Infinity)
4    00:10:46.737 Poison squashed: 10.23.0.0/24 lost if
5    00:10:46.737 Poison squashed: 10.23.0.0/24 rt net gone
6    00:10:46.737 Route installing: 10.23.0.0/24 192.168.3.1
7    00:10:46.737 Send reply: 10.23.0.0/24 192.168.3.1
8    00:10:46.737 Not active net/1=SH: 10.23.0.0/24 1
9    00:10:46.737 FC not sat Dmin/met: metric(Infinity) metric(1782272)
10   00:10:46.737 Find FS: 10.23.0.0/24 metric(1782272)
11   00:10:46.737 Rcv query met/succ met: metric(Infinity) metric(Infinity)
12   00:10:46.737 Rcv query dest/nh: 10.23.0.0/24 192.168.3.1
```

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b. Issue the **no shutdown** command on R2 interface G0/0/1.

c. Configure D1 as an EIGRP stub router.

```
D1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)# router eigrp 98
D1(config-router)# eigrp stub
D1(config-router)# end
D1#
Mar  6 00:11:40.624: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 192.168.3.1
(GigabitEthernet1/0/11) is down: peer info changed
Mar  6 00:11:45.174: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 192.168.3.1
(GigabitEthernet1/0/11) is up: new adjacency
```

d. Verify that R1 sees switch D1 as a stub by examining the output of the **show ip eigrp neighbor detail** command.

```
R1# show ip eigrp neighbor detail
EIGRP-IPv4 Neighbors for AS(98)
H   Address                 Interface              Hold Uptime   SRTT   RTO  QSeq
                                                   (sec)         (ms)       CntNum
2   192.168.3.2             Gi0/0/1                  14 00:00:39    3   100  010
   Version 25.0/2.0, Retrans: 1, Retries: 0, Prefixes: 3
   Topology-ids from peer - 0
   Topologies advertised to peer:   base

   Stub Peer Advertising (CONNECTED SUMMARY ) Routes
   Suppressing queries
1   10.13.0.3               Se0/1/0                  11 00:11:44    1   100  028
   Time since Restart 00:04:06
   Version 25.0/2.0, Retrans: 0, Retries: 0, Prefixes: 3
   Topology-ids from peer - 0
   Topologies advertised to peer:   base

0   10.12.0.2               Gi0/0/0                  12 00:12:24    1   100  030
   Time since Restart 00:04:07
   Version 25.0/2.0, Retrans: 2, Retries: 0, Prefixes: 2
   Topology-ids from peer - 0
   Topologies advertised to peer:   base

Max Nbrs: 0, Current Nbrs: 0
```

e. Issue the **shutdown** command on R2 interface G0/0/1. Take note of the timestamp on the syslog message reporting that the interface is down.

f. On switch D1, issue the **show ip eigrp events** command. You will see that no query was received looking for the 10.23.0.0/24 network. The query was stopped at R1, speeding up the convergence process.

```
D1# show ip eigrp events
Event information for AS 98:
1    00:12:53.776 NDB delete: 10.23.0.0/24 1
2    00:12:53.776 Poison squashed: 10.23.0.0/24 rt net gone
```

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```
3    00:12:53.776 RDB delete: 10.23.0.0/24 192.168.3.1
4    00:12:53.776 Not active net/1=SH: 10.23.0.0/24 1
5    00:12:53.776 FC not sat Dmin/met: metric(Infinity) metric(1782272)
6    00:12:53.776 Find FS: 10.23.0.0/24 metric(1782272)
7    00:12:53.776 Rcv update met/succmet: metric(Infinity) metric(Infinity)
8    00:12:53.776 Rcv update dest/nh: 10.23.0.0/24 192.168.3.1
9    00:12:53.776 Ignored route, hopcount: 10.23.0.0 255
10   00:12:50.077 Metric set: 172.16.3.0/24 metric(1910016)
11   00:12:50.077 Route installed: 172.16.3.0/24 192.168.3.1
12   00:12:50.077 Route installing: 172.16.3.0/24 192.168.3.1
13   00:12:50.077 Find FS: 172.16.3.0/24 metric(Infinity)
14   00:12:50.077 Free reply status: 172.16.3.0/24
15   00:12:50.077 Clr handle num/bits: 0 0x0
16   00:12:50.077 Clr handle dest/cnt: 172.16.3.0/24 0
17   00:12:50.077 Rcv reply met/succ met: metric(1910016) metric(1909760)
18   00:12:50.077 Rcv reply dest/nh: 172.16.3.0/24 192.168.3.1
19   00:12:50.077 Metric set: 10.23.0.0/24 metric(1782272)
20   00:12:50.076 Route installed: 10.23.0.0/24 192.168.3.1
21   00:12:50.076 Route installing: 10.23.0.0/24 192.168.3.1
22   00:12:50.076 Find FS: 10.23.0.0/24 metric(Infinity)
23   00:12:50.076 Free reply status: 10.23.0.0/24
```

g. Issue the **no shutdown** command on R2 interface G0/0/1.

Step 4: Filter EIGRP routes with a distribute list.

EIGRP supports several different filtering capabilities. The simplest and most direct is to use a distribute list. A distribute list refers to an access list which can be applied to all EIGRP updates being sent by a certain router, or it can be applied to a specific interface to modify updates as they exit. For this exercise, we will filter the 10.12.0.0/24 network from updates being sent out of R2 interface G0/0/1. This will cause a change in R3's routing table.

a. Examine the routing table on R3 by issuing the **show ip route eigrp | begin Gateway** command. In the output, you can see that R3 has calculated the path via R2 at 10.23.0.2 to be the best path to reach the 10.12.0.0/24 network.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D        10.12.0.0/24 [90/3072] via 10.23.0.2, 00:00:17, GigabitEthernet0/0/0
D     192.168.0.0/22 [90/3328] via 10.23.0.2, 00:00:53, GigabitEthernet0/0/0
```

b. Our intent is to change the configuration at R2 so that R3 only learns about the 10.12.0.0/24 network from R1. Create an access list that denies the 10.12.0.0/24 network and permits all other networks.

```
R2# config t
R2(config)# ip access-list standard EIGRP-FILTER
R2(config-std-nacl)# deny 10.12.0.0 0.0.255.255
R2(config-std-nacl)# permit any
R2(config-std-nacl)# exit
```

c. Enter EIGRP router configuration mode and configure the distribute list to reference the access list you just created, further specifying that the filter should be effective outbound on interface G0/0/1.

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```
R2(config)# router eigrp 98
R2(config-router)# distribute-list EIGRP-FILTER out g0/0/1
R2(config-router)# end
*Mar 6 00:19:56.379: %DUAL-5-NBRCHANGE: EIGRP-IPv4 98: Neighbor 10.23.0.3
(GigabitEthernet0/0/1) is resync: intf route configuration changed
```

d. On R3, issue the **show ip route eigrp | begin Gateway** command. As you can see, the successor for the 10.12.0.0/24 network has changed to R1 at 10.13.0.1.

```
R3# show ip route eigrp | begin Gateway
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
D        10.12.0.0/24 [90/1792256] via 10.13.0.1, 00:01:30, Serial0/1/0
D     192.168.0.0/22 [90/3328] via 10.23.0.2, 00:03:00, GigabitEthernet0/0/0
```

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